📄 Healthcare Admissions & Revenue Analytics

End-to-End Microsoft Fabric Project Documentation

🧭 1. Project Overview

This project focuses on building an **end-to-end analytics solution** using Microsoft Fabric and Power BI to analyze healthcare admissions and revenue trends.

The objective was to:

- Track key KPIs such as admissions, revenue, and patient count
- Identify trends, seasonal patterns, and performance variations
- Deliver insights through interactive dashboards

🎯 2. Objectives

- Build a scalable analytics pipeline using Microsoft Fabric
- Design a structured data model using star schema
- Perform time-based trend analysis
- Identify peak and low admission periods
- Enable data-driven decision-making through dashboards

🏗️ 3. Architecture & Tools Used

🔹 Tools & Technologies

- Microsoft Fabric
- Dataflows Gen2
- Lakehouse (for data storage)

- Power BI (for reporting)
- DAX (for calculations and analysis)

🔹 Architecture Flow

Data Source → Dataflows Gen2 → Lakehouse → Semantic Model → Power BI Dashboard

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🔄 4. Data Ingestion & Transformation

🔹 Dataflows Gen2

- Used for data ingestion and transformation
- Performed:
    - Data cleaning
    - Column creation (Year, Month, YearMonthDate)
    - Data type standardization

🔹 Key Transformations

- Created **YearMonthDate** for time-based analysis
- Derived:
    - Year
    - Month
    - Month Name

🧱 5. Data Modeling

🔹 Schema Design

- Implemented **Star Schema**

🔹 Tables

Fact Table:

- `fact_admissions`
    - Admission Date

    - Revenue
    - Patient Count
    - Medical Condition
    - Doctor ID
    - Hospital ID

Dimension Tables:

- `dim_date`
- `dim_doctor`
- `dim_hospital`
- `dim_medical_condition`

🔹 Relationships

- One-to-Many relationships between dimensions and fact table
- Single-direction filtering for optimized performance

📊 6. KPI Development

🔹 Key Metrics Created

- Total Admissions
- Total Revenue
- Total Patients
- Average Revenue per Patient

📈 7. Time Intelligence (DAX)

🔹 Concepts Applied

- Monthly trend analysis
- Time-based aggregation
- Handling continuous date axis

🔹 Key Learning

- Time intelligence works only with **proper date columns**
- Importance of **granularity (monthly vs daily)**

🔍 8. Advanced Analysis

🔹 Peak & Dip Detection

Implemented dynamic measures:

- Identified **maximum (peak) values**
- Identified **minimum (dip) values**

🔹 Logic Used

- `MAXX()` and `MINX()` with filter context
- Compared current value with overall max/min

🔹 Outcome

- Automatically highlights:
  - 🔺 Highest admission period
  - 🔻 Lowest admission period

📊 9. Dashboard Design

🔹 Visuals Created

- KPI Cards (Revenue, Patients, Admissions)
- Monthly Admissions Trend (Line Chart)
- Peak & Dip Highlight Chart
- Supporting breakdown visuals

🔹 Design Principles Applied

- Clean layout (minimal clutter)
- Consistent color theme
- Business-friendly titles
- Insight-driven storytelling

💡 10. Key Insights

- Admissions remained relatively stable over time
- Seasonal fluctuations were observed
- Clear identification of peak and low periods
- Operational trends can be monitored effectively

🚧 11. Challenges Faced

🔹 1. Time Intelligence Confusion

- Difficulty understanding MoM and date context
- Learned importance of proper date hierarchy

🔹 2. Continuous vs Categorical Axis

- Continuous axis caused formatting issues
- Learned trade-offs between analytics vs presentation

🔹 3. Tooltip Date Formatting

- Default tooltip showed DateTime (00:00:00)
- Learned workaround using formatted columns

🔹 4. Dynamic Peak/Dip Labeling

- Initially used manual shapes (static)
- Replaced with dynamic DAX measures

🔹 5. Fabric UI Limitations

- Fewer formatting options compared to Power BI Desktop
- Learned workaround strategies

🧠 12. Key Learnings

🔹 Technical Learnings

- End-to-end pipeline in Microsoft Fabric
- Dataflows Gen2 usage
- Star schema modeling
- DAX fundamentals and context handling
- Time intelligence concepts

🔹 Analytical Learnings

- Importance of data granularity
- Identifying trends vs anomalies
- Converting raw data into insights

🔹 Visualization Learnings

- Importance of clean design
- Avoiding clutter
- Highlighting key data points (peak/dip)

🚀 13. Future Enhancements

- Add MoM / YoY analysis
- Implement dynamic insight cards
- Add drill-through for deeper analysis
- Enhance tooltip with contextual insights
- Integrate real-time data

🏁 14. Conclusion

This project demonstrates the ability to:

- Build an end-to-end analytics solution using Microsoft Fabric
- Perform data modeling and transformation
- Apply DAX for advanced analysis
- Design business-focused dashboards

It reflects a strong foundation in:
👉 Data analytics
👉 Business intelligence
👉 Data storytelling